Amirhossein Imani

Senior AI Engineer

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SUMMARY

AI engineer with 4 years of experience building and shipping LLM systems and the production infrastructure behind them. Work on an LLM agent platform (LangGraph, FastAPI) that turns natural-language requests into validated, self-repairing code, streamed live to users. Own the execution backbone: Kubernetes job orchestration, usage metering, observability, and guardrails. Hands-on with retrieval-augmented generation (RAG), prompt engineering, and structured LLM outputs. Previously owned the full model lifecycle (distributed search, validation, deployment, drift monitoring) of a production ML platform.

EXPERIENCE

Roshan AI, *AI Engineer* 02/2026 – Present | Tehran

Sharkdev is an LLM agent platform where users describe what they want in natural language (~24 languages), and a LangGraph agent extracts a structured spec, generates code, validates it through real Kubernetes runs, and self-heals failures in an LLM repair loop (up to 5 iterations).

- Extended the LangGraph code-generation agent by adding brokerage resolution to spec-extraction and generation prompts and integrating live-runtime helpers into generation and repair prompts, enabling generated code to work with real external systems, not just tests.
- Built the workflow orchestration service executed by the agent, including job submission to Kubernetes workers via NATS JetStream.
- Built the live-execution module end to end, including session lifecycle management, encrypted secret storage, asynchronous background handling, and WebSocket/SSE streaming to the UI; converted the service layer to async to increase concurrency.
- Built Pilotfish from scratch, an asynchronous observability service with NATS event tracking, live Kubernetes pod monitoring, and Prometheus metrics for validation jobs.
- Built the production layer of Ahab, a parallel optimization engine, including Kubernetes worker orchestration, Redis job management, webhooks, Prometheus metrics, and stale-worker cleanup; contributed to its core, reducing a 679,140-point parameter search from ~88 hours to under 3 minutes.
- Developed the runtime package for agent-generated code, including in-code LLM calls and structured LLM output parsing.

Roshan AI, *Machine Learning Engineer* 10/2022 – 01/2026 | Tehran

*Owned the mo*del lifecycle infrastructure of a production ML platform, from hyperparameter search to live deployment and monitoring.

- Moved hyperparameter search from a single server to Kubernetes: dockerized the workers and designed the workflow (10 pods, each running 5 Optuna TPE workers with different seeds on shared MySQL study state), then automated search, training, validation and selection to run from a single commit ID.
- Designed and built an automated out-of-sample validation stage that re-scores every candidate on untouched data using the production code path, closing the gap where models looked strong in search but degraded in forward performance.
- Designed and built the selection process that turns thousands of trials into deployable models: tree-based selection, decision-tree analysis of what separates winning trials, rolling-window evaluation, and HTML/JSON reports

- Defined the evaluation methodology: rolling validation and selection windows, randomly sampled search slices, and comparison of selection methods against random-selection baselines across ten rolling offsets.
- Led performance engineering with cProfile and memory profilers: fixed a cross-validation memory leak by restructuring per-fold cache lifetimes (pattern reused in four more places), removed a dataframe memory blowup, and replaced pandas hot paths with NumPy.
- Built a live monitoring suite (signal drift, distribution, frequency, outcome reports, health checks) with Grafana dashboards, plus model-exporter support for automatic deployment.

EDUCATION

Shahid Beheshti University, *bachelor of computer engineering* 09/2017 – 09/2022 | Tehran

SKILLS

LLM / GenAI (LLM agents, tool calling, multi-provider LLM fallback, LangGraph, RAG, prompt engineering, structured outputs , guardrails and validation, LLM-assisted code generation, SSE/WebSocket streaming) |
ML (PyTorch, Optuna; hyperparameter optimization, cross-validation, model validation) |
Backend / MLOps (Python, FastAPI,, Kubernetes, Docker, NATS JetStream, Redis, PostgreSQL, Prometheus, Grafana)